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1. Authentication & Registration – User registration and login (Rules R1, R2, R3, R4)

- YCondition is true / applicable
- NCondition is false / not satisfied
- Not applicable / irrelevant for the rule
- TAction is triggered

Type	Condition / Action	R1	R2	R3	R4
	Rule / Scenario	-	-	N	N
CONDITIONS					
C	Authenticated Session	-	-	N	N
C	Registration Requested	Y	Y	N	-
C	Registration Data Valid	Y	Y	-	-
C	Username Unique	Y	N	-	-
ACTIONS					
A	Create user account	T			
A	Reject account creation		T		
A	Validate credentials			T	
A	Issue JWT session token			T	
A	Grant dashboard access			T	
A	Deny dashboard access				T
A	Redirect to login page				T

2. Dashboard & Navigation – Dashboard display and folder navigation (Rules R5, R6, R7, R8, R9, R10)

- YCondition is true / applicable
- NCondition is false / not satisfied
- Not applicable / irrelevant for the rule
- TAction is triggered

Type	Condition / Action	R5	R6	R7	R8	R9	R10
	Rule / Scenario	Y	Y	Y	Y	Y	Y
CONDITIONS							
C	Authenticated Session	Y	Y	Y	Y	Y	Y
C	Registration Requested	N	N	N	N	N	N
C	Folder Valid/Selected	Y	Y	Y	Y	Y	Y
C	File Selected	N	N	N	N	N	N
C	Metadata Indexed	-	Y	-	-	-	-
C	File Metadata Exists	N	Y	-	-	-	-
C	Resource Belongs to User	Y	Y	Y	Y	Y	Y
C	Folder Empty	Y	N	-	-	-	-
ACTIONS							
A	Query current user's root folder	T					
A	Display empty dashboard placeholder	T					
A	Query by parent folder and user		T				
A	Render matching folders/files		T				
A	Create root-level virtual folder			T			
A	Refresh dashboard			T			
A	Set current folder				T		
A	Filter by Parent_Folder_ID				T		
A	Render selected directory contents				T		
A	Create child folder under selected parent		Telegram Backed	Hybrid Cloud Storage	Management System	T	

A	Refresh folder contents					T	
A	Move UI state to parent folder						T
A	Refresh parent contents						T

3. Upload & Telegram Transfer – File upload and Telegram transfer handling (Rules R11, R12, R13, R14, R15, R16, R17, R18)

- YCondition is true / applicable
- NCondition is false / not satisfied
- Not applicable / irrelevant for the rule
- TAction is triggered

Type	Condition / Action	R11	R12	R13	R14	R15	R16	R17	R18
	Rule / Scenario	Y	Y	Y	Y	Y	Y	Y	Y
CONDITIONS									
C	Authenticated Session	Y	Y	Y	Y	Y	Y	Y	Y
C	Registration Requested	N	N	N	N	N	N	N	N
C	Folder Valid/Selected	Y	Y	Y	Y	Y	Y	Y	Y
C	File Selected	Y	Y	Y	Y	Y	Y	Y	Y
C	File ≤ 50 MB	Y	N	N	-	-	-	-	-
C	File ≤ 2 GB	Y	Y	N	Y	Y	Y	Y	Y
C	Network Available	Y	Y	-	N	Y	Y	Y	Y
C	Telegram Transfer Successful	Y	Y	-	N	N	N	Y	Y
C	HTTP 429	N	N	-	N	Y	N	N	N
C	Telegram file_id Received	Y	Y	-	N	N	N	N	Y
C	Metadata Indexed	Y	Y	-	N	N	N	N	N
C	File Metadata Exists	N	N	-	N	N	N	N	N
C	Resource Belongs to User	Y	Y	Y	Y	Y	Y	Y	Y
ACTIONS									
A	Stream upload	T							
A	Send non-chunked transfer within 50 MB ceiling	T							
A	Use streaming/chunking fallback		T						
A	Forward chunks without persistent local copy		T						
A	Capture Telegram file_id	T	T						
A	Write metadata to SQL	T	T						
A	Make file visible after indexing	T	T						
A	Update upload progress	T							
A	Reject upload			T					
A	Display file-size/limit error			T					
A	Stop/handle transfer gracefully				T				
A	Display explicit network/transfer failure message				T				
A	Do not index incomplete upload				T				
A	Pause for designated cool-down					T			
A	Retry/continue transfer according to implementation					T			
A	Keep application running					T			
A	Do not index until transfer succeeds					T			
A	Fail transfer gracefully						T		
A	Display controlled transfer/API error						T		
A	Do not index file						T		
A	Do not index incomplete metadata							T	
A	Report upload/indexing failure							T	
A	Keep file unavailable in dashboard								T
A	Report metadata/indexing failure								T
A	Preserve required metadata consistency								T

4. Download & Access – File download and cross-user access control (Rules R19, R20, R21)

- YCondition is true / applicable
- NCondition is false / not satisfied
- Not applicable / irrelevant for the rule
- TAction is triggered

Type	Condition / Action	R19	R20	R21
	Rule / Scenario	Y	Y	Y
CONDITIONS				
C	Authenticated Session	Y	Y	Y
C	Registration Requested	N	N	N
C	File Selected	Y	Y	Y
C	File ≤ 2 GB	Y	-	-
C	Network Available	Y	-	-
C	Telegram file_id Received	Y	N	Y
C	Metadata Indexed	Y	N	Y
C	File Metadata Exists	Y	N	N
C	Resource Belongs to User	Y	Y	N

ACTIONS				
A	Look up relational metadata	T		
A	Retrieve remote data using file_id	T		
A	Stream binary response to browser	T		
A	Reject download request		T	
A	Display file-not-found/metadata error		T	
A	Deny resource access			T
A	Do not expose Telegram file_id			T
A	Preserve user isolation			T

5. Delete Operations – Deleting files and folders (Rules R22, R23, R24)

Y	Condition is true / applicable
N	Condition is false / not satisfied
-	Not applicable / irrelevant for the rule
T	Action is triggered

Type	Condition / Action	R22	R23	R24
	Rule / Scenario	Y	Y	Y
CONDITIONS				
C	Authenticated Session	Y	Y	Y
C	Registration Requested	N	N	N
C	Folder Valid/Selected	-	Y	Y
C	File Selected	Y	N	N
C	Telegram file_id Received	Y	-	-
C	Metadata Indexed	Y	-	-
C	File Metadata Exists	Y	-	-
C	Resource Belongs to User	Y	Y	Y
C	Folder Empty	-	Y	N
ACTIONS				
A	Refresh dashboard	T		
A	Delete SQL metadata record	T		
A	Unlink file from virtual folder tree	T		
A	Delete virtual folder		T	
A	Display folder-not-empty/operation error			T

6. Search – Global search behaviour (Rules R25, R26)

Y	Condition is true / applicable
N	Condition is false / not satisfied
-	Not applicable / irrelevant for the rule
T	Action is triggered

Type	Condition / Action	R25	R26
	Rule / Scenario	Y	Y
CONDITIONS			
C	Authenticated Session	Y	Y
C	Registration Requested	N	N
C	File Selected	N	N
C	Metadata Indexed	Y	-
C	File Metadata Exists	Y	N
C	Resource Belongs to User	Y	Y
C	Search Text Entered	Y	Y
ACTIONS			
A	Query metadata across user's hierarchy	T	
A	Return matching files	T	
A	Preserve folder hierarchy	T	
A	Return no matching records		T
A	Display empty search result state		T

7. Administrative Export – Administrative metadata export (Rules R27, R28)

Y	Condition is true / applicable
N	Condition is false / not satisfied
-	Not applicable / irrelevant for the rule
T	Action is triggered

Type	Condition / Action	R27	R28
	Rule / Scenario	Y	Y
CONDITIONS			
C	Authenticated Session	Y	Y
C	Registration Requested	N	N
C	File Selected	N	N
C	Admin Authorized	Y	N
ACTIONS			

A	Verify administrative permission	T	
A	Export SQL metadata mapping/backup	T	
A	Return backup to administrator	T	
A	Deny administrative export		T
A	Do not expose metadata backup		T

8. UI Preferences – Theme toggle / appearance (Rules R29)

- Y

Condition is true / applicable
- N

Condition is false / not satisfied
- Not applicable / irrelevant for the rule
- T

Action is triggered

Type	Condition / Action	R29
	Rule / Scenario	Y
CONDITIONS		
C	Authenticated Session	Y
C	Registration Requested	N
C	File Selected	N
C	Resource Belongs to User	Y
ACTIONS		
A	Toggle global stylesheet between light/dark mode	T